



ORIGINAL RESEARCH ARTICLE

Digital Media, Genetics, and Risk for ADHD Symptoms in Children: A Longitudinal Study

Samson Nivins, PhD,¹ Michael A. Mooney, PhD,² Joel Nigg, PhD,³ Torkel Klingberg, PhD¹

ABSTRACT

BACKGROUND: Children spend significant amounts of time using digital media (DM), and longer exposure may increase attention-deficit/hyperactivity disorder (ADHD)-related symptoms, although findings are mixed. We investigated longitudinal association between different types of DM use and ADHD-related symptoms in school-aged children, accounting for genetic predisposition and socioeconomic status.

METHODS: This study included children from the Adolescent Brain Cognitive Development Study, and these children were followed annually for 4 years. Estimated time spent on social media, video games, and television/videos was self-reported using Youth Screen Time Survey. ADHD-related symptoms were assessed at each visit with the parent-reported Child Behavior Checklist. Genetic predisposition was estimated using a polygenic risk score for ADHD (PGS-ADHD).

RESULTS: The study included 8324 children (53% boys; mean age: 9.9 years). On average, children spent 2.3 hours/d watching television/videos, 1.4 hours/d on social media, and 1.5 hours/d playing video games. Average social media use was associated with increased inattention symptoms over time (β [SE], 0.03 [0.01]; $P < .001$), with a cumulative 4-year effect of $\beta = 0.15$ (SE, 0.03; $P < .001$). No associations were found between playing video games or watching television/videos and ADHD-related symptoms. The association between social media use and inattention symptoms was not moderated by sex, ADHD diagnosis, PGS-ADHD, or ADHD medication status. Inattention symptoms were not associated with increased social media use over time.

CONCLUSION: Social media use was associated with an increase in inattention symptoms in children over time. Although the observed effect size was small, it could have significant consequences if behavior changes occur at the population level.



¹Department of Neuroscience, Karolinska Institute, Stockholm, Sweden

²Division of Bioinformatics and Computational Biology, Oregon Health and Science University, Portland, Oregon

³Division of Clinical Psychology, Oregon Health and Science University, Portland, Oregon

Address correspondence to: Torkel Klingberg, PhD, Department of Neuroscience, Karolinska Institutet, Stockholm 17165, Sweden. torkel.klingberg@ki.se

Dr Nivins had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Dr Nivins conceptualized and designed the study, carried out the formal analysis, drafted the initial manuscript, and critically reviewed and revised the manuscript. Dr Klingberg conceptualized and designed the study, drafted the manuscript, and critically reviewed and revised the manuscript. Drs Mooney and Nigg computed the polygenic scores for attention-deficit/hyperactivity disorder and critically reviewed and revised the manuscript. The data used for the analyses presented in this paper are from the Adolescent Brain Cognitive Development (ABCD) Study (<https://abcdstudy.org>; National Institute of Mental Health Data Archive [NDA]). Data can be accessed by directly applying to the NDA.

CONFLICT OF INTEREST DISCLOSURES: The authors have no conflicts of interest relevant to this article to disclose.

WHAT'S KNOWN ON THIS SUBJECT: Attention-deficit/hyperactivity disorder (ADHD) is a common, highly heritable neurodevelopmental disorder. Its rising prevalence suggests environmental influences. Digital media use has been associated with ADHD symptoms, although findings remain mixed. Most research is cross-sectional and lacks focus on specific media or genetic predisposition.

WHAT THIS STUDY ADDS: Social media use, unlike other digital media activities, was associated with increased risk of inattention symptoms. Given its widespread use among children, the findings underscore the need for stricter age regulations and consideration of platform design to support healthy development.

Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a childhood-onset neurodevelopmental disorder with a heritability estimated between 60% and 80%.^{1–3} Despite this high heritability, the prevalence of ADHD diagnoses in children has risen significantly. According to US National Survey of Children's Health, parent-reported ADHD prevalence increased from 9.5% in 2003 to 2007 to 11.3% in 2020 to 2022.^{4,5} This rise may reflect past underdiagnosis, greater public awareness, current overdiagnosis, or impact of environmental factors. Multiple environmental factors including maternal stress, lead exposure, and perinatal factors can contribute to symptomatology.⁶ Further, there could also be an interaction between genetic predisposition and environmental exposures.^{7,8}

An environmental factor that lately has drawn increasing interest is the potential negative impact of using digital media (DM), including social media (eg, Facebook or Instagram), playing video games, or watching television/videos. A meta-analysis of longitudinal and cross-sectional studies from 1987 to 2011 found a small but statistically significant association between DM use and ADHD-related symptoms (effect size [*b*], 0.12).⁹ However, cross-sectional studies are subjected to self-selection bias, meaning that ADHD symptoms may lead to increased DM use,¹⁰ rather than the other way around. Among longitudinal studies, digital multitasking was associated with increased inattention symptoms in younger adolescents (*b*, 0.16), but not in older.¹¹ Another study also found that DM use in adolescents predicted later ADHD symptoms (odds ratio [OR], 1.11).¹² Watching television and playing video games showed a weak association with hyperactivity symptoms 2 years later in children aged 10 years (*b*, 0.04), but not in children aged 6 years.¹³ Conversely, a study in children aged 10 years found no significant association between DM use and inattention symptoms 6 to 12 months later when correcting for baseline symptomatology.¹⁴

In summary, there are indications of associations between different types of DM use and ADHD symptoms, but the literature is inconsistent.¹⁵ Differences in study outcomes could be due to differences in the definition of "Screen Time," different lengths of follow-up, and differences among study populations. None of these studies accounted for genetic predisposition, which could explain differences in outcomes if populations differ systematically. To our knowledge, one study found that children with a high-genetic predisposition for ADHD measured using polygenic risk scores (PGS-ADHD) were more likely to spend increased time using DM.¹⁶ It is theoretically possible that children with higher PGS-ADHD scores may be more sensitive to DM exposure, which could trigger or exacerbate ADHD symptoms.

The present study aimed to examine longitudinal associations between different types of DM use and ADHD-related symptoms in school-aged children, incorporating genetic

predisposition to ADHD through PGS-ADHD. We hypothesized that, if DM use influences ADHD symptoms, children with above-average use exhibit a greater increase in ADHD symptoms over time.

Methods

Study Design and Participants

The behavioral data were obtained from the Adolescent Brain Cognitive Development (ABCD) Study (<https://abcdstudy.org/>; data release 5.0),¹⁷ a longitudinal cohort comprising 11 875 children born between 2005 and 2009. These children were enrolled at ages 9 to 10 years from 21 data collection sites across the United States between 2016 and 2018.¹⁸ Children were excluded: if they were born extremely preterm (<28 weeks'), had a birthweight of less than 1200 g, were not proficient in English, had any neurological problems, had a history of seizures, or had a contraindication to undergo brain magnetic resonance imaging scans. We included only 1 child per family at random to mitigate familial-related effects in the analysis, leaving 8324 children for the baseline (T_0) analysis. All children and their parents/guardians provided informed written assent/consent for participation, and the central Institutional Review Board at the University of California, San Diego approved the study protocols.¹⁹ This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

Children were enrolled at ages 9 to 10 years (baseline, T_0) and followed annually for up to 4 years (T_1 – T_4), with follow-up rates exceeding 85% through the third year and 43% at year 4. Detailed information on ADHD-related symptoms and the DM use survey are provided in Methods in the Supplemental Material. Brief descriptions are provided below.

ADHD-Related Symptoms

ADHD-related symptoms were assessed at all study visits using the parent-reported Child Behavior Checklist (CBCL), which consists of 118 items rated on a 3-point Likert scale (0 = not true, 1 = somewhat or sometimes true, and 2 = very true or often true). We used combined ADHD-related symptoms as continuous variables based on the *Diagnostic and Statistical Manual of Mental Disorders (DSM)-5*-defined ADHD CBCL subscale constructed by experts (ie, *cbcl_scr_dsm5_adhd*), with higher scores indicating greater ADHD symptomatology. Coefficient alpha was satisfactory at all time points, with values of α being greater than 0.72.

To examine specific-symptom domains, ADHD symptoms were further categorized into "inattention" and "hyperactivity-impulsivity," based on *DSM-5* criteria, using CBCL items only. The 9 CBCL items (questions) corresponding to each domain were analyzed in a factor analysis.²⁰ Items with a factor loading larger than 0.3 were retained,²¹ resulting in 4 inattention items and 5

hyperactivity/impulsivity items (Methods in Supplemental Material). Although CBCL includes a broader set of questions labeled “inattention,” our factor-derived inattention score was highly correlated with the CBCL summary inattention score at T_0 ($r, 0.93$). Higher scores on specific-symptom domain scores indicate greater severity of inattention or hyperactivity-impulsivity symptoms.

Exposure DM Use

The estimated time spent on social media use, playing video games, or watching television/videos was assessed using the Youth Screen Time Survey across multiple visits: T_0 , T_1 , T_2 , T_3 , and T_4 . Children provided reports on the number of hours they spent on weekdays and weekends excluding school-related work on the following: (1) watching television or movies; (2) watching videos (eg, YouTube); (3) playing video games on a computer, console, phone, or another device (eg, Xbox); (4) texting on a cell phone, tablet, or computer (eg, Google Chat, WhatsApp); (5) visiting social networking sites (eg, Facebook, Instagram); and (6) using video chat (eg, Skype, FaceTime). DM use was categorized based on our previous publications,^{22,23} as follows: (a) social media use (4 + 5 + 6), (b) playing video games (3), or (c) watching television/videos (1 + 2). The average daily hours spent on individual DM use were calculated as [(total weekday hours $\times$ 5) + (total weekend hours $\times$ 2)]/7.

The Youth Screen Time Survey asks children to report hours spent on each activity for a “typical” weekday and a “typical” weekend day, rather than for a specific prior day, week, or month.²⁴ This approach is intended to capture habitual patterns of media use, reflecting each child’s average daily behavior on usual days rather than relying on short-term recall over a fixed reference period.

We relied on self-reported surveys by children,^{25,26} considering that parents might not have full awareness, especially among children aged 9 to 10 years and older who frequently use DM unsupervised.

Due to COVID-19 lockdown, children might likely spend more time using DM than anticipated at T_0 . A US-based study reported a 2-fold increase in DM use during lockdown.²⁷ To account for this change, we used the average estimated time spent on individual DM category (ie, social media use, playing video games, and watching television/videos) across visits for longitudinal analyses, rather than relying solely on T_0 or T_4 . Each category was analyzed separately in longitudinal models.

Genotyping

Saliva samples were collected at T_0 visit, and genotyping was performed at Rutgers University Cell and DNA Repository using Affymetrix NIDA SmokeScreen array, as previously described.^{28,29}

Processed genotypes were obtained from National Institute of Mental Health Data Archive (<https://dx.doi.org/10.15154/1503209>), and standard quality control procedures were applied. Samples and single nucleotide polymorphisms (SNPs) with low call rates were excluded, and imputation was conducted using IMPUTE2 with 1000 Genomes (phase 3) reference panel after prephasing with SHAPEIT.^{30,31} Only those SNPs imputed with high confidence (INFO > 0.8) were retained.

Polygenic scores for ADHD (PGS-ADHD) were derived from the 2016 to 2017 PGC + iPSYCH ADHD GWAS meta-analysis (20 183 cases, 35 191 controls) using the LDpred method.^{32,33} The PGS-ADHD score was standardized (mean, 0; SD, 1) and included in all models along with the first 10 genetic principal components (PCs) to account for potential genetic confounding (Methods in Supplemental Material).

Covariates

All covariates were selected a priori based on known associations with DM use and/or ADHD-related symptoms.^{34,35} Analyses included age at T_0 visit, sex assigned at birth, socioeconomic status (SES), study sites, PGS-ADHD, and the first 10 genetic PCs. Age was considered because DM use typically increases as children grow older.³⁶ Sex was included given well-documented differences in both DM use and ADHD prevalence.³⁷ SES, derived from parental education, household income, and neighborhood quality using PC analysis, was included since lower SES has been associated with higher DM use.³⁸ Study site was included to account for site-specific variations. Age and sex were obtained from caregiver-reported developmental history questionnaires (Methods in Supplemental Material).

Statistical Analysis

Characteristics are presented as frequencies and percentages for categorical variables and as mean with SDs for continuous variables. Prior to main analyses, associations between PGS-ADHD, individual DM use at T_0 , and ADHD-related symptoms at T_0 were examined using Pearson correlation.

Longitudinal associations between average DM use and ADHD-related symptoms were analyzed using separate linear mixed models with random intercept and slopes for each DM category, examining combined and individual presentations of inattention and hyperactivity-impulsivity symptoms.³⁹ Fixed effects included DM use, time, age at T_0 , sex, SES, PGS-ADHD, and the first 10 genetic PCs, with study sites as random effects. Two-way interaction (DM use $\times$ time) captured longitudinal effects (ie, between-person) and three-way interaction (DM use $\times$ time $\times$ PGS-ADHD) tested whether genetic predisposition moderated these effects. Models were fitted using ‘lmer’ function in lme4 package with restricted maximum likelihood. Statistical significance was set at a P value of less than 0.05, with Bonferroni correction for multiple comparisons ($0.05/9 = 0.005$).

Standardized yearly effect sizes of β greater than 0.05 or cumulative β greater than 0.10 were considered meaningful.^{40–42}

For significant longitudinal associations ($P < 0.005$, uncorrected), moderation models were fitted separately to test DM use $\times$ time interactions individually by sex, ADHD diagnosis at T_0 , and ADHD medication status at T_0 . ADHD diagnosis and medication use were determined from caregiver reports using the Kiddie Schedule for Affective Disorders and Schizophrenia (KSADS) and modified Medication Inventory survey. Cumulative effects of DM use on symptom change were estimated by regressing total change in ADHD-related symptoms ($T_4 - T_0$) on average DM use, adjusting for predefined covariates (Methods in Supplemental Material).

To assess the robustness of association between social media use and inattention symptoms, we conducted 5 additional analyses. These included (1) limiting the cohort to children born at-term (≥ 37 weeks') to control for preterm birth and (2) restricting to typically developing children by excluding those with ADHD and co-occurring comorbidities (ie, intellectual disabilities, conduct disorder, oppositional defiant disorder, or generalized anxiety disorder) identified via National Institutes of Health Toolbox WISC-V and caregiver KSADS reports. In a previous ABCD study, approximately 3.5% of children met criteria for ADHD and approximately 70% had at least 1 comorbidity.²⁸ Excluding these children reduced diagnostic heterogeneity and tested whether the observed negative association extended to children without underlying neurodevelopmental or psychiatric diagnoses; (3) including only children with behavioral data at all time points; (4) restricting analysis to T_3 due to lower T_4 follow-up; and (5) exploring whether average inattention symptoms predicted longer social media use (Methods in Supplemental Material).

We also conducted Cross-Lagged Panel Models to confirm the directionality of associations between social media use and inattention symptoms (Methods in Supplemental Material).

All analyses were conducted using R version 3.5 (RStudio).

Results

Sample Characteristics

Of the 11 875 children recruited into the ABCD study cohort, 8324 children (mean [SD] age, 9.9 [0.6]; boys, n [%] = 4408 [53.0]) fulfilled our inclusion criteria at T_0 (Table 1). Throughout 4 follow-up waves, children spent an average of 1.4 hours/d on social media, 1.5 hours/d playing video games, and 2.3 hours/d watching television/videos. The cohort was ethnically diverse: white, 4356 (52.3%); Black, 1358 (16.3%); Asian, 496 (6.0%); Pacific Islander, 2 (0.02%); Native American, 183 (2.2%); Hispanic, 1763 (21.2%); and Other, 106 (1.3%). PGS-ADHD was positively associated with both ADHD symptoms and DM use at T_0 (Figure 1).

TABLE 1. Characteristics of the Study Population (N = 8324)

Characteristics	Overall Cohort
Age at T_0 , mean (SD), years	9.9 (0.6)
Sex, male, n (%)	4408 (53.0)
Race or ethnicity, n (%)	
White	4356 (52.3)
Black	1358 (16.3)
Asian	496 (6.0)
Pacific Islander	2 (0.02)
Native American	183 (2.2)
Hispanic	1763 (21.2)
Other	106 (1.3)
Maternal education, n (%)	
High school or less	121 (1.5)
High school	411 (4.9)
High school graduate	833 (10.0)
Bachelor's degree	2369 (28.5)
Some college/associate degree	2413 (29.0)
Master's degree	1639 (19.7)
Professional degree	528 (6.3)
Parental income, n (%)	
$\leq 49\,999$	2206 (26.5)
50 000–74 999	1056 (12.7)
75 000–99 999	1132 (13.6)
100 000–199 999	2359 (28.3)
$\geq 200\,000$	886 (10.6)
Socioeconomic status, mean (SD)	0.03 (1.4)
Raw polygenic risk score for ADHD, mean (SD)	−7.5 (0.3)
Social media use (hours/day), mean (SD)	
T_0	0.5 (1.1)
T_1	0.9 (1.6)
T_2	1.3 (1.4)
T_3	1.9 (1.5)
T_4	2.5 (1.4)
Playing video games (hours/day), mean (SD)	
T_0	1.0 (1.1)
T_1	1.2 (1.2)
T_2	1.5 (1.4)
T_3	1.9 (1.5)
T_4	2.0 (1.6)
Watching television/videos (hours/day), mean (SD)	
T_0	2.2 (1.8)
T_1	2.4 (1.8)
T_2	2.3 (1.3)

(Continued on next page)

TABLE 1. Characteristics of the Study Population (N = 8324) (Continued)

Characteristics	Overall Cohort
Total digital media use by children at T_0 (hours/day), mean SD	3.7 (3.0)
Average estimated time spent by children over 4 years (hours/day), mean (SD)	
Social media use	1.4 (1.0)
Playing video games	1.5 (1.0)
Watching television/videos	2.3 (1.3)
Total digital media use by children as reported by parents at T_0 (hours/day)	3.0 (2.4)
ADHD stimulant medications, n (%)	778 (0.9)
ADHD-related symptoms at T_0 , mean (SD)	
Combined ADHD	2.7 (3.0)
Inattention	1.7 (1.9)
Hyperactivity-impulsivity	1.3 (1.8)

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; N, total study population; n, subgroup of total population; T_0 , baseline; T_1 , one-year follow-up; T_2 , 2-year follow-up; T_3 , 3-year follow-up; T_4 , 4-year follow-up.

DM Use and ADHD-Related Symptoms

The average use of playing video games (video games $\times$ time: β [SE], -0.05 [0.01]; $P < .001$) or watching television/videos (watching television/videos $\times$ time: β [SE], -0.05 [0.01]; $P < .001$)

was associated with a decrease in hyperactivity-impulsivity symptoms over time (Table 2). In contrast, the average use of social media use was associated with an increase in inattention symptoms over time (social media $\times$ time: β [SE], 0.03 [0.01]; $P < .001$). To illustrate this effect, average social media use was divided into 4 quartiles and plotted against inattention symptoms over time (Figure 2A). A significant three-way interaction was observed between social media use, time, and PGS-ADHD with combined ADHD symptoms (social media $\times$ time $\times$ PGS-ADHD: β [SE], -0.04 [0.01]; $P = .002$). This indicates that the association between social media use and changes in combined ADHD symptoms over time differed depending on PGS-ADHD.

Additional variables, including sex, ADHD diagnosis, or medication status at T_0 , did not moderate the association between DM use and ADHD-related symptoms over time (Supplemental Tables 1–3).

Cumulative Effect Size of DM Use on ADHD-Related Symptoms (T_4 - T_0)

Average social media use was positively associated with an increase in inattention symptoms (β [SE], 0.15 [0.03]; $P < .001$) over the study period. To illustrate this effect, average social media use was divided into 4 quartiles and plotted against changes in inattention symptoms (Figure 2B). In contrast, neither the association between average time spent watching

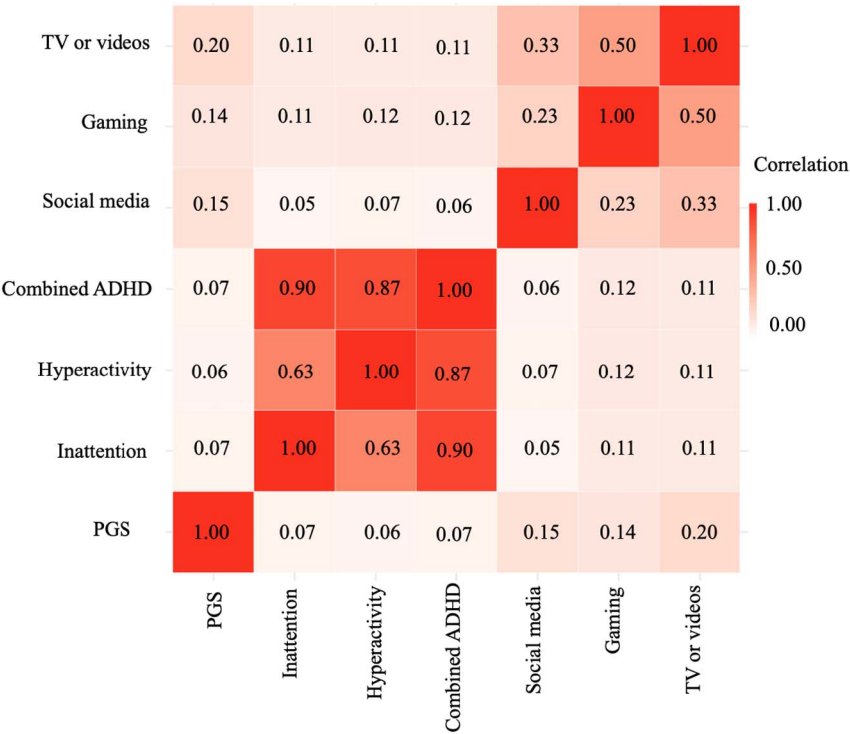


FIGURE 1. Association between polygenic scores, digital media use at T_0 , and attention-deficit/hyperactivity disorder symptoms at T_0 . Abbreviations: ADHD, attention-deficit/hyperactivity disorder; PGS, polygenic risk scores; TV, television.

TABLE 2. Association Between Digital Media Use Over 4 Years and Attention-Deficit/Hyperactivity Disorder–Related Symptoms in the Overall Cohort (N = 8324)

Symptoms	Social Media Use	SES	PGS-ADHD	Social Media Use × Time	Social Media Use × Time × PGS-ADHD
Combined ADHD	0.11 (0.04)	−0.22 (0.04) ^a	0.54 (0.05) ^a	0.03 (0.01)	−0.04 (0.01) ^b
Inattention	0.05 (0.02)	−0.14 (0.02) ^a	0.29 (0.02) ^a	0.03 (0.01) ^a	−0.01 (0.01)
Hyperactivity-impulsivity	0.21 (0.04) ^a	−0.25 (0.04) ^a	0.50 (0.05) ^a	−0.03 (0.01)	−0.01 (0.01)
	Playing Video Games	SES	PGS-ADHD	Playing Video Games × Time	Playing Video Games × Time × PGS-ADHD
Combined ADHD	0.30 (0.04) ^a	−0.17 (0.04) ^a	0.51 (0.05) ^a	−0.01 (0.01)	−0.01 (0.01)
Inattention	0.21 (0.02) ^a	−0.11 (0.02) ^a	0.27 (0.02) ^a	−0.01 (0.01)	−0.01 (0.01)
Hyperactivity-impulsivity	0.33 (0.04) ^a	−0.22 (0.04) ^a	0.48 (0.05) ^a	−0.05 (0.01) ^a	−0.01 (0.01)
	Watching Television or Videos	SES	PGS-ADHD	Watching Television or Videos × Time	Watching Television or Videos × Time × PGS-ADHD
Combined ADHD	0.39 (0.04) ^a	−0.14 (0.04) ^b	0.49 (0.05) ^a	−0.02 (0.01)	−0.02 (0.01)
Inattention	0.23 (0.02) ^a	−0.10 (0.02) ^a	0.26 (0.02) ^a	−0.01 (0.01)	−0.01 (0.01)
Hyperactivity-impulsivity	0.46 (0.04) ^a	−0.17 (0.04) ^a	0.46 (0.05) ^a	−0.05 (0.01) ^a	−0.01 (0.01)

Data are presented as standardized beta (standard error).
Abbreviations: ADHD, attention-deficit/hyperactivity disorder; N, total study population; PGS, polygenic risk scores; SES, socioeconomic status.
^a Significance level of <0.001 (Bonferroni corrected).
^b Significance level of <0.005 (Bonferroni corrected).

television/videos and total change in hyperactivity-impulsivity symptoms (β [SE], −0.06 [0.02]; $P = .004$), nor that between playing video games and total change in hyperactivity-impulsivity symptom change (β [SE], −0.05 [0.03]; $P = .06$), met our predefined threshold of $\beta = 0.10$ for a population or clinically relevant cumulative effect size.

Additional Analyses

Additional analyses supported the robustness of association between social media use and inattention. Restricting the sample to children born at-term or to typically developing children yielded consistent effect sizes, indicating results were

not driven by preterm birth or pre-existing conditions. Including only children with complete behavioral data or limiting follow-up to 3 years did not alter results. Potential recursive effects of inattention on later social media use were statistically significant but negligible (β [SE], −0.01 [0.001]; $P < .001$; total effect ($T_4 - T_0$) = −0.01) (Figure 2C) (Supplemental Tables 4–8).

Cross-lagged panel models confirmed a unidirectional association of social media use predicted increases in inattention symptom over time (β [SE], 0.03 [0.01]; $P = .004$), whereas inattention did not predict increased social media use (β [SE], −0.002 [0.004]; $P = .64$) (Figure 3).

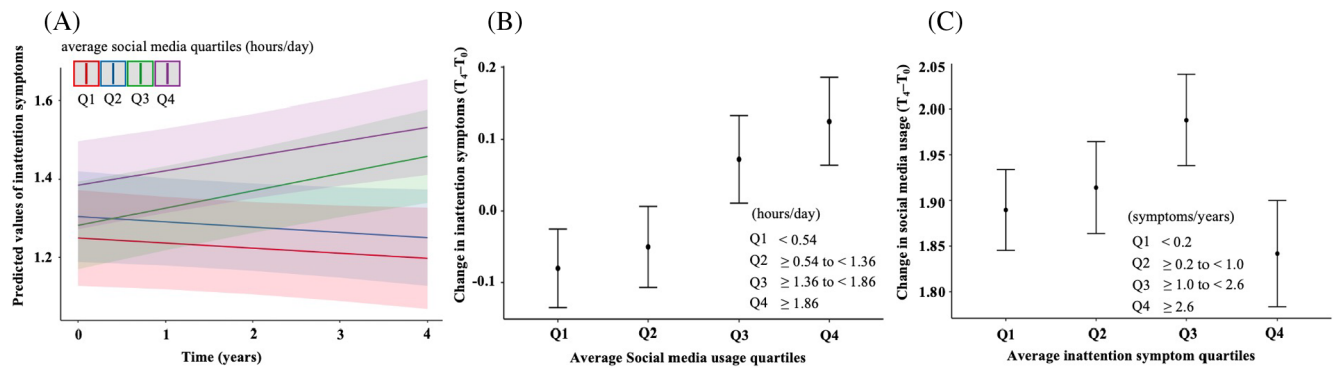


FIGURE 2. (A) Average social media use and inattention symptoms plotted over time, where average social media use (hours/d) was segregated into different quartiles (Q1, 0.61; Q2, ≥ 0.61 to < 1.22 ; Q3, ≥ 1.22 to < 1.97 ; Q4, ≥ 1.97). Children in the highest quartiles of average social media use showed a significant increase in inattention symptom scores over time; (B) change in inattention symptom scores over 4 years, plotted for each quartile of average social media use; and (C) change in social media use over 4 years, plotted for each quartile for average inattention symptoms.

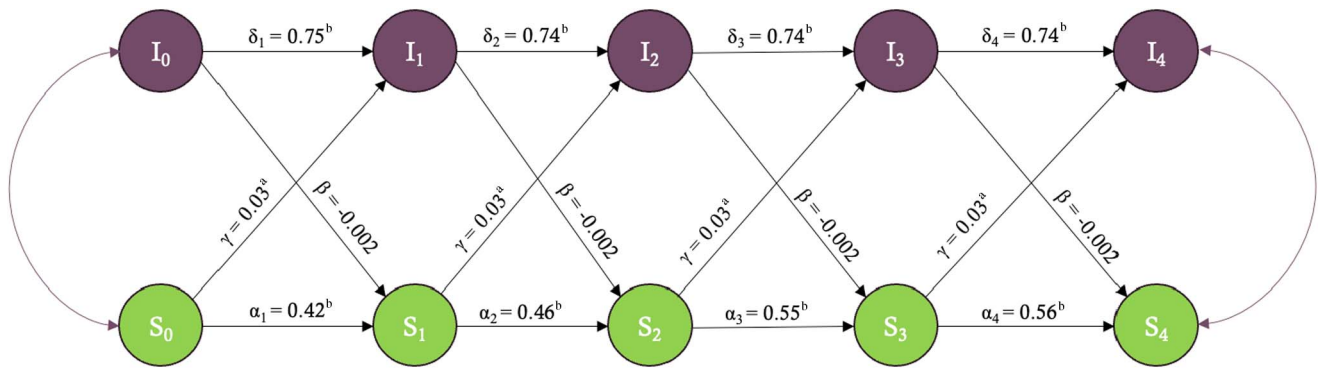


FIGURE 3. Cross-lagged panel models showing associations between social media use and inattention symptoms. All coefficients are standardized β weights.

Abbreviations: I, inattention symptoms; S, social media use. 0 to 4 represent the follow-up time points. ^a $P < .01$. ^b $P < .001$.

Discussion

We investigated the long-term association between different types of DM use and changes in ADHD-related symptoms over 4 years. Consistent with our hypothesis, there was a positive interaction between average social media use and time, such that children with above-average social media use showed a greater increase in inattention symptoms compared with others. This interaction was specific to social media and not observed for either watching television/videos or playing video games. The associations remained robust across all additional analyses, indicating that findings were not driven by specific subgroups. Importantly, there was no evidence of reverse association, as average inattention symptoms did not predict increased social media use (Figure 2C), and cross-lagged panel models further supported one-directional relationship, with higher social media use preceding increases in inattention symptoms (Figure 3). Together, these results strengthen the interpretation of a potentially causal link between social media use and changes in inattention symptoms.

In this study, social media use was specifically associated with a gradual increase in inattention symptoms over 4 years. Although yearly effect size was small (Table 2), cumulative effect over the study period was 0.15 (Figure 2B), showing a linear increase over time (Figure 2A). This finding aligns with prior research showing associations between digital multitasking or DM use and later inattention symptoms with comparable effect sizes ($d = 0.16$ and $OR = 1.11$, respectively).^{11,12} Importantly, the association persisted after accounting for genetic predisposition and point to a higher degree of specificity, as similar patterns were not observed for playing video games or watching television/videos.

Although genetic liability for ADHD explained a significant portion of average ADHD symptomatology, it did not moderate the association between DM use and symptom change. Surprisingly, PGS-ADHD scores were strongly correlated with DM use (Figure 1), which may partly explain the cross-sectional

associations between DM use and ADHD-related symptoms.

However, despite this shared genetic influence across all forms of DM use, only social media use was prospectively associated with gradual increase in inattention symptoms.

An effect size of 0.15 has small consequences for an individual's risk of meeting diagnostic criteria and likely minimal functional impact in daily life. However, at population level, such an effect could have substantial consequences. For example, assume a diagnosis of ADHD-inattentive type is met when inattention supersedes 1.65SD above population mean, corresponding to 5% of children. If social media use increases by 1SD across the population, a conservative estimate of behavioral change over the past decade and is associated with 0.15SD increase in inattention, the proportion of children exceeding this threshold would increase by approximately 35%. This relative increase translates to prevalence rising from 5% to approximately 6.8%. Using a more recent prevalence estimate of 11.3%, the same 35% increase would imply a rise to approximately 14.4%. Although hypothetical, these calculations illustrate how even small average shifts in symptoms can have meaningful consequences at population level.

We can only speculate about the mechanisms underlying the association between social media use and increased inattention symptoms. Social media platforms often involve constant messaging and notifications, which can disrupt attention and interfere with current activities. Experimental studies have shown that such interruptions, or even the mere presence of a mobile phone nearby without using it, can impair attention and learning on psychological tests.⁴³ This can be contrasted with playing video games, which requires sustained attention. The results of both experimental and longitudinal studies^{44,45} reveal that paying video games has been associated with improvements in cognitive function.

We found a steady increase in social media use from approximately 0.5 hours/d at age 9 years to 2.5 hours by age 13 years, despite most platforms, such as Facebook and TikTok,

requiring users to be aged 13 years or older. This early and increasing social media use underscores the need for stricter age verification and clearer guidelines for tech companies. Policymakers should reinforce regulations to limit access for younger children and ensure platforms are age appropriate to support healthy development.

This study has several strengths, including its large sample size, longitudinal design, inclusion of genetic information, and detailed categorization of DM use. However, some limitations should be noted. DM use was self-reported rather than objectively measured. Although self-reporting by children and adolescents of DM use is likely more accurate than parent reporting, it remains possible that children with more ADHD symptoms may misestimate their DM use, as difficulties with time estimation are well-known in ADHD. To address this, we conducted an additional analysis excluding children with ADHD and co-occurring comorbidities, and the findings persisted, suggesting robustness. Further, detailed information on specific activities, such as which games were played, which social media platforms were used, and the time of day, would have added valuable insight. Measuring screen time alone may be insufficient, as recent research suggests that the impact of DM depends more on the nature of engagement than on duration. ADHD assessments relied on caregiver/parent reports from the KSADS and CBCL; teacher reports, which are often recommended for a more comprehensive evaluation, were not consistently available. Similarly, information on ADHD medication use was only available at T_0 and not tracked longitudinally, and families were not systematically asked about professional ADHD diagnoses across follow-ups. Finally, because ABCD study excluded children with sensory or major neurological conditions, severe intellectual disabilities, moderate-to-severe autism spectrum disorder, or extremely preterm birth, the findings may not be generalizable to these populations. Despite these limitations, it is unlikely that they would introduce systematic errors large enough to explain the observed association between social media use and inattention symptoms, or the differing trends seen between social media and gaming.

Conclusion

We identified an association between social media use and increased inattention symptoms, interpreted here as a likely causal effect. Although the effect size is small at an individual level, it could have significant consequences if behavior changes across population level. These findings suggest that social media use may contribute to rising incidence of ADHD diagnoses. Future research should investigate the underlying mechanisms, examine the impact of specific social media behaviors, and evaluate strategies to mitigate potential risks.

Acknowledgments

The authors are grateful to the ABCD study members and their families for their participation. Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive Development (ABCD) Study (<https://abcdstudy.org>), held in the National Institute of Mental Health Data Archive. This is a multisite, longitudinal study designed to recruit more than 10 000 children aged 9 and 10 years and follow them over 10 years into early adulthood. The ABCD Study^{*} is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123, and U24DA041147. A full list of supporters is available at <https://abcdstudy.org/federal-partners.html>. A listing of participating sites and a complete listing of the study investigators can be found at https://abcdstudy.org/consortium_members/. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or ABCD consortium investigators. The ABCD data repository grows and changes over time. The ABCD data used in this report came from the ABCD Data Release 5.0 (data accessed 2023; DAR ID: 14933). We also would like to thank Professor Ulrika Ådén for providing feedback on the manuscript.

FUNDING: This study was supported by the Swedish Research Council to Dr Klingberg; and Stiftelsen Frimurare Barnhuset, post-doc grant to Dr Nivins. None of those listed here had any part in data handling, data analysis, or result interpretation.

Accepted for Publication Date: November 19, 2025

<https://doi.org/10.1542/pedsos.2025-000922>

This is an open access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License which permits noncommercial distribution and reproduction in any medium, provided the original author and source are credited. © 2026 Nivins S et al. *Pediatrics Open Science*.

References

- Chen Q, Brikell I, Lichtenstein P, et al. Familial aggregation of attention-deficit/hyperactivity disorder. *J Child Psychol Psychiatry*. 2017;58(3):231–239. PubMed doi: 10.1111/jcpp.12616
- Grimm O, Kittel-Schneider S, Reif A. Recent developments in the genetics of attention-deficit hyperactivity disorder. *Psychiatry Clin Neurosci*. 2018;72(9):654–672. PubMed doi: 10.1111/pcn.12673
- Chang Z, Lichtenstein P, Asherson PJ, Larsson H. Developmental twin study of attention problems: high heritabilities throughout development. *JAMA Psychiatry*. 2013;70(3):311–318. PubMed doi: 10.1001/jamapsychiatry.2013.287
- Reuben C, Elgaddal N. Attention-deficit/hyperactivity disorder in children ages 5–17 years: United States, 2020–2022. *NCHS Data Brief*. 2024;499:1–9. PubMed doi: 10.15620/CDC/148043
- Visser SN, Danielson ML, Bitsko RH, et al. Trends in the parent-report of health care provider-diagnosed and medicated attention-deficit/hyperactivity disorder: United States, 2003–2011. *J Am Acad Child Adolesc Psychiatry*. 2014;53(1):34–46.e2. PubMed doi: 10.1016/j.jaac.2013.09.001
- Kim JH, Kim JY, Lee J, et al. Environmental risk factors, protective factors, and peripheral biomarkers for ADHD: an umbrella review. *Lancet Psychiatry*. 2020;7(11):955–970. PubMed doi: 10.1016/S2215-0366(20)30312-6
- Nigg J, Nikolas M, Burt SA. Measured gene-by-environment interaction in relation to attention-deficit/hyperactivity disorder. *J Am Acad Child Adolesc Psychiatry*. 2010;49(9):863–873. PubMed doi: 10.1016/j.jaac.2010.01.025
- Martel MM, Nikolas M, Jernigan K, Friderici K, Waldman I, Nigg JT. The dopamine receptor D4 gene (DRD4) moderates family environmental effects on ADHD. *J Abnorm Child Psychol*. 2011;39(1):1–10. PubMed doi: 10.1007/s10802-010-9439-5
- Nikkelen SW, Valkenburg PM, Huizinga M, Bushman BJ. Media use and ADHD-related behaviors in children and adolescents: a meta-analysis. *Dev Psychol*. 2014;50(9):2228–2241. PubMed doi: 10.1037/a0037318
- Masi L, Abadie P, Herba C, Emond M, Gingras MP, Amor LB. Video games in ADHD and non-ADHD children: modalities of use and association with ADHD symptoms. *Front Pediatr*. 2021;9:632272. PubMed doi: 10.3389/fped.2021.632272
- Baumgartner SE, van der Schuur WA, Lemmens JS, te Poel F. The relationship between media multitasking and attention problems in adolescents: results of two longitudinal studies. *Hum Commun Res*. 2018;44(1):3–30. doi: 10.1093/hcre.12111
- Ra CK, Cho J, Stone MD, et al. Association of digital media use with subsequent symptoms of attention-deficit/hyperactivity disorder among adolescents. *JAMA*. 2018;320(3):255–263. PubMed doi: 10.1001/jama.2018.8931
- Allen MS, Vella SA. Screen-based sedentary behaviour and psychosocial well-being in childhood: cross-sectional and longitudinal associations. *Ment Health Phys Act*. 2015;9:41–47. doi: 10.1016/j.mhpa.2015.10.002
- Barlett ND, Gentile DA, Barlett CP, Eisenmann JC, Walsh DA. Sleep as a mediator of screen time effects on US children's health outcomes: a prospective study. *J Child Media*. 2012;6(1):37–50. doi: 10.1080/17482798.2011.633404
- Thorell LB, Burén J, Ström Wiman J, Sandberg D, Nutley SB. Longitudinal associations between digital media use and ADHD symptoms in children and adolescents: a systematic literature review. *Eur Child Adolesc Psychiatry*. 2022;33(8):2503–2526. PubMed
- Yang A, Rolls ET, Dong G, et al. Longer screen time utilization is associated with the polygenic risk for attention-deficit/hyperactivity disorder with mediation by brain white matter microstructure. *EBioMedicine*. 2022;80:104039. PubMed doi: 10.1016/j.ebiom.2022.104039
- Adolescent Brain Cognitive Development (ABCD) Study. Accessed August 27, 2023. Updated 2023. <https://abcdstudy.org>.
- Casey BJ, Cannonier T, Conley MI, et al. The Adolescent Brain Cognitive Development (ABCD) study: imaging acquisition across 21 sites. *Dev Cogn Neurosci*. 2018;32:43–54. doi: 10.1016/j.dcn.2018.03.001
- Auchter AM, Hernandez Mejia M, Heyser CJ, et al. A description of the ABCD organizational structure and communication framework. *Dev Cogn Neurosci*. 2018;32:8–15. doi: 10.1016/j.dcn.2018.04.003
- Bauermeister JJ, Barkley RA, Bauermeister JA, Martínez JV, McBurnett K. Validity of the sluggish cognitive tempo, inattention, and hyperactivity symptom dimensions: neuropsychological and psychosocial correlates. *J Abnorm Child Psychol*. 2012;40(5):683–697. PubMed doi: 10.1007/s10802-011-9602-7
- Hair Jr JF, Black WC, Babin BJ, Anderson RE. Multivariate data analysis. *Multivariate Data Analysis*. Pearson; 2010:785–785.
- Sauce B, Liebherr M, Judd N, Klingberg T. The impact of digital media on children's intelligence while controlling for genetic differences in cognition and socioeconomic background. *Sci Rep*. 2022;12(1):7720. PubMed doi: 10.1038/s41598-022-11341-2
- Nivins S, Sauce B, Liebherr M, Judd N, Klingberg T. Long-term impact of digital media on brain development in children. *Scientific Reports*. 2024;14(1):13030. doi: 10.1038/s41598-024-63566-y
- Bagot KS, Tomko RL, Marshall AT, et al. Youth screen use in the ABCD[®] study. *Developmental Cognitive Neuroscience*. 2022;57:101150. doi: 10.1016/j.dcn.2022.101150
- Brener ND, Billy JO, Grady WR. Assessment of factors affecting the validity of self-reported health-risk behavior among adolescents: evidence from the scientific literature.

- J Adolesc Health*. 2003;33(6):436–457. PubMed doi: 10.1016/S1054-139X(03)00052-1
26. Riley AW. Evidence that school-age children can self-report on their health. *Ambul Pediatr*. 2004;4(4)(suppl):371–376. PubMed doi: 10.1367/A03-178R.1
27. Nagata JM, Cortez CA, Cattle CJ, et al. Screen time use among US adolescents during the COVID-19 pandemic: findings from the Adolescent Brain Cognitive Development (ABCD) study. *JAMA Pediatr*. 2022;176(1):94–96. PubMed doi: 10.1001/jamapediatrics.2021.4334
28. Cordova MM, Antovich DM, Ryabinin P, et al. Attention-deficit/hyperactivity disorder: restricted phenotypes prevalence, comorbidity, and polygenic risk sensitivity in the ABCD baseline cohort. *J Am Acad Child Adolesc Psychiatry*. 2022;61(10):1273–1284. PubMed doi: 10.1016/j.jaac.2022.03.030
29. Baurley JW, Edlund CK, Pardamean CI, Conti DV, Bergen AW. Smokescreen: a targeted genotyping array for addiction research. *BMC Genomics*. 2016;17(128):145. PubMed doi: 10.1186/s12864-016-2495-7
30. Howie BN, Donnelly P, Marchini J. A flexible and accurate genotype imputation method for the next generation of genome-wide association studies. *PLoS Genet*. 2009;5(6):e1000529. PubMed doi: 10.1371/journal.pgen.1000529
31. Delaneau O, Zagury J-F, Marchini J. Improved whole-chromosome phasing for disease and population genetic studies. *Nat Methods*. 2013;10(1):5–6. PubMed doi: 10.1038/nmeth.2307
32. Demontis D, Walters RK, Martin J, et al; ADHD Working Group of the Psychiatric Genomics Consortium (PGC); Early Lifecourse & Genetic Epidemiology (EAGLE) Consortium; 23andMe Research Team. Discovery of the first genome-wide significant risk loci for attention deficit/hyperactivity disorder. *Nat Genet*. 2019;51(1):63–75. PubMed doi: 10.1038/s41588-018-0269-7
33. Vilhjálmsson BJ, Yang J, Finucane HK, et al; Schizophrenia Working Group of the Psychiatric Genomics Consortium, Discovery, Biology, and Risk of Inherited Variants in Breast Cancer (DRIVE) study. Modeling linkage disequilibrium increases accuracy of polygenic risk scores. *Am J Hum Genet*. 2015;97(4):576–592. PubMed doi: 10.1016/j.ajhg.2015.09.001
34. Green A, Baroud E, DiSalvo M, Faraone SV, Biederman J. Examining the impact of ADHD polygenic risk scores on ADHD and associated outcomes: a systematic review and meta-analysis. *J Psychiatr Res*. 2022;155:49–67. PubMed doi: 10.1016/j.jpsychires.2022.07.032
35. Russell AE, Ford T, Williams R, Russell G. The association between socioeconomic disadvantage and attention deficit/hyperactivity disorder (ADHD): a systematic review. *Child Psychiatry Hum Dev*. 2016;47(3):440–458. PubMed doi: 10.1007/s10578-015-0578-3
36. Sina E, Buck C, Veidebaum T, et al. Media use trajectories and risk of metabolic syndrome in European children and adolescents: the IDEFICS/I.Family cohort. *International Journal of Behavioral Nutrition and Physical Activity*. 2021;18(1):134. doi: 10.1186/s12966-021-01186-9
37. Twenge JM, Martin GN. Gender differences in associations between digital media use and psychological well-being: evidence from three large datasets. *J Adolesc*. 2020;79(1):91–102. PubMed doi: 10.1016/j.adolescence.2019.12.018
38. Ke Y, Chen S, Hong J, Liang Y, Liu Y. Associations between socioeconomic status and screen time among children and adolescents in China: a cross-sectional study. *PLoS One*. 2023;18(3):e0280248. PubMed doi: 10.1371/journal.pone.0280248
39. Cnaan A, Laird NM, Slasor P. Using the general linear mixed model to analyse unbalanced repeated measures and longitudinal data. *Stat Med*. 1997;16(20):2349–2380. PubMed doi: 10.1002/(SICI)1097-0258(19971030)16:20<2349:AID-SIM667>3.0.CO;2-E
40. Kraft MA. Interpreting effect sizes of education interventions. *Educ Res*. 2020;49(4):241–253. doi: 10.3102/0013189X20912798
41. Abelson RP. A variance explanation paradox: when a little is a lot. *Psychol Bull*. 1985;97(1):129–133. doi: 10.1037/0033-2909.97.1.129
42. Pogrow S. How effect size (practical significance) misleads clinical practice: The case for switching to practical benefit to assess applied research findings. *The American Statistician*. 2019;73(suppl1):223–234.
43. Van Der Schuur WA, Baumgartner SE, Sumter SR, Valkenburg PM. The consequences of media multitasking for youth: a review. *Comput Human Behav*. 2015;53:204–215. doi: 10.1016/j.chb.2015.06.035
44. Bediou B, Adams DM, Mayer RE, Tipton E, Green CS, Bavelier D. Meta-analysis of action video game impact on perceptual, attentional, and cognitive skills. *Psychol Bull*. 2018;144(1):77–110. PubMed doi: 10.1037/bul0000130
45. Sauce B, Liebherr M, Judd N, Klingberg T. The impact of digital media on children's intelligence while controlling for genetic differences in cognition and socioeconomic background. *Scientific Reports*. 2022;12(1):7720. doi: 10.1038/s41598-022-11341-2.